

Safe Interval Motion Planning for Quadrotors in Dynamic Environments

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| Outline

Central research question

RESEARCH QUESTION

How can a quadrotor reason about when and how to pass through a dynamic environment without searching the full space-time state space?

Introduction

Method

Experiment

Conclusion

| Introduction

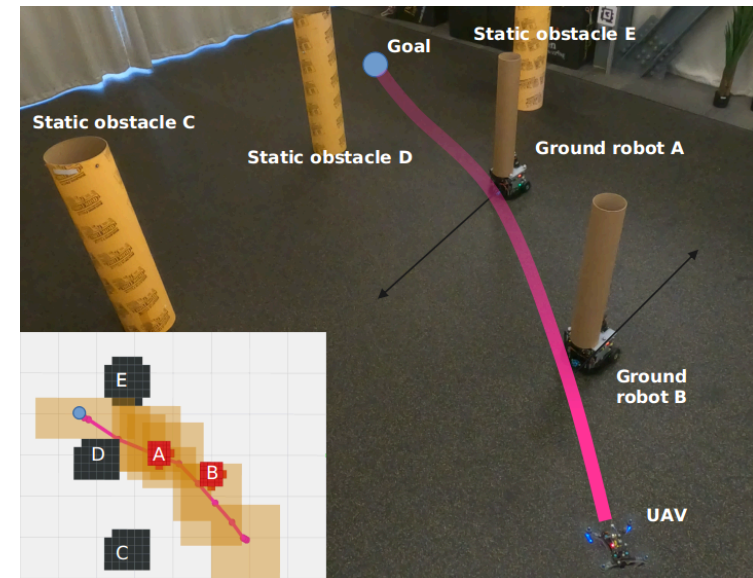
Known Dynamic Obstacles: Space and Time Are Coupled

Problem setting

- 3D quadrotor navigation with static + moving obstacles
- Obstacle trajectories known in finite horizon $T = [t_s, t_e]$
- Goal: safe, dynamically feasible, and smooth trajectory

Why existing methods fall short

- **SIPP and SIPP-IP**: grid-cell-level intervals; dense-3D pre-computation costly; ignores quadrotor dynamics
- **T-PRM and VIS-PRM**: roadmap front ends reduce space, but temporal safety still depends on vertex time stamps, edge checks, or short edges
- **UVD**: useful for topological paths, but assumes **equal temporal domains** and fails when start and end times differ



edge-safe

intervals

temporal

topology

B-spline

smooth

Method

Edge Safe Intervals: When Can the Quadrotor Traverse?

Collision intervals

$$CI(e) = \bigcup_i (t_i, t_{i+1})$$

times when edge e intersects any moving obstacle X_{obs}^t

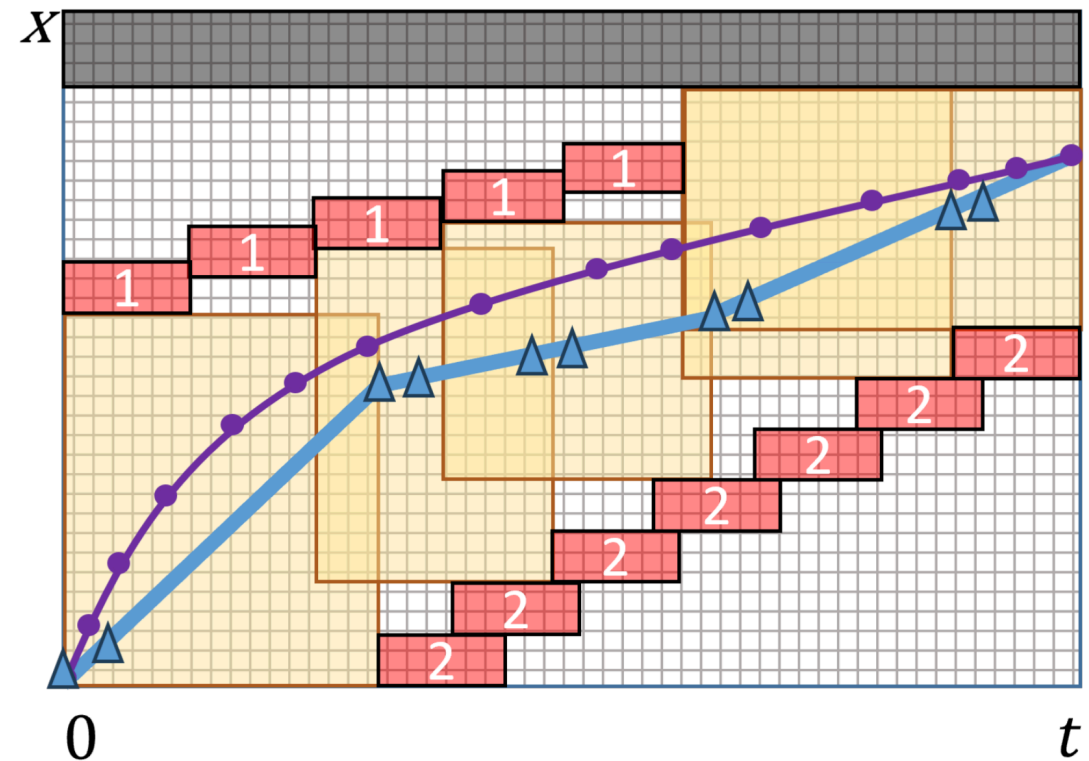
Safe intervals

$$SI(e) = \bigcup_j (t_j, t_{j+1}), \quad t_{j+1} - t_j > t_{\min}$$

- $(t_j, t_{j+1}) \in T \setminus CI(e)$, filtered by $t_{\min}$
- $t_{\min}$: derived from trapezoidal velocity profile (speed/accel limits)

Edge-level safe $\subseteq$ vertex-level safe

two safe endpoints $\nRightarrow$ safe edge interior



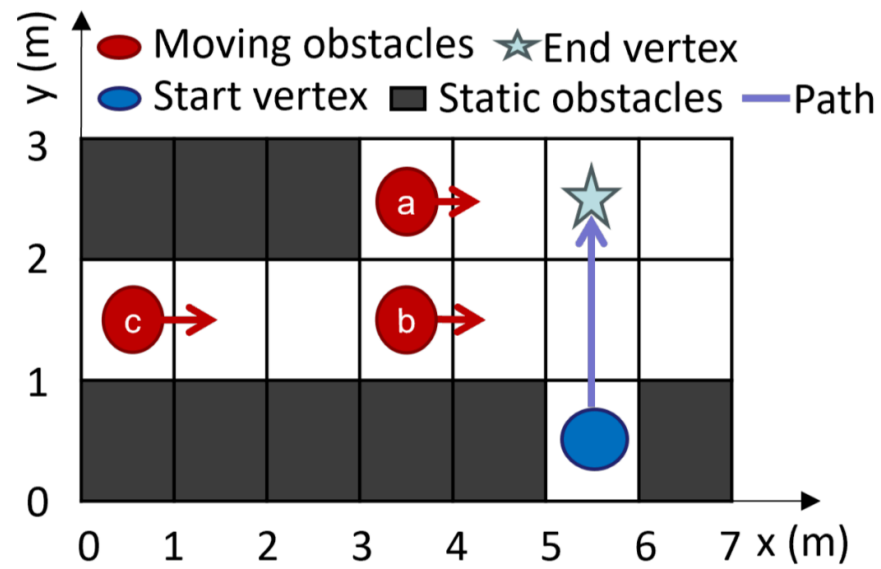
Method

Temporal Corridor and UTVD: Chaining and Classifying

Temporal corridor

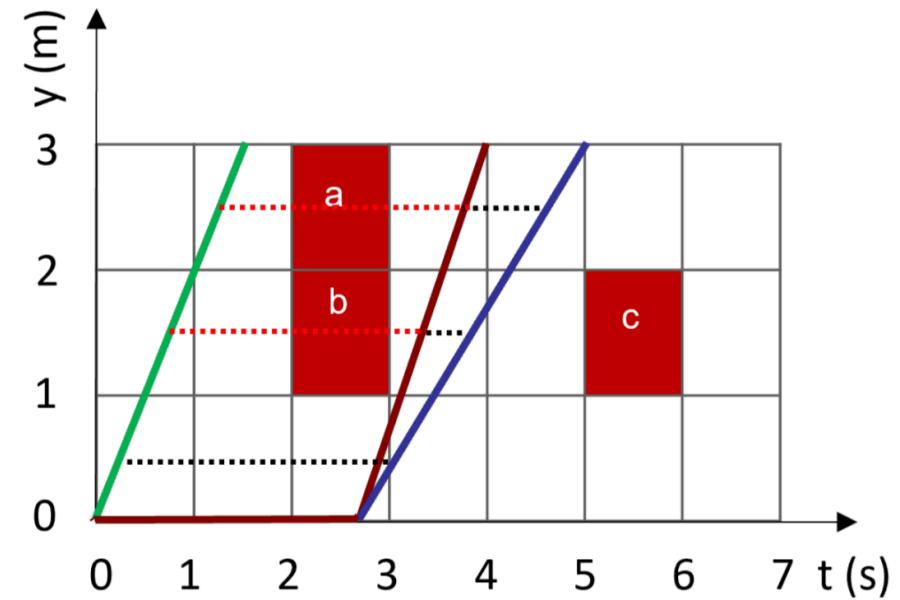
$$TC(\pi) = \{I_k \mid I_k \subseteq SI(e_k) \cap SI(e_{k+1})\}$$

- adjacent SI must **overlap** — relay must connect
- **directed**: $\rightarrow \pi \not\leftarrow \pi$



$$\text{UTVD: } s' = \alpha s + \theta, \quad \alpha \in \mathbb{R}^+, \theta \in \mathbb{R}$$

- same class iff $\overline{\sigma_1(s), \sigma_2(\alpha s + \theta)}$ collision-free $\forall s$
- static UVD ($\alpha = 1, \theta = 0$) fails with unequal time domains



Method

Dynamic Connected Visibility Graph

Guards and Connectors

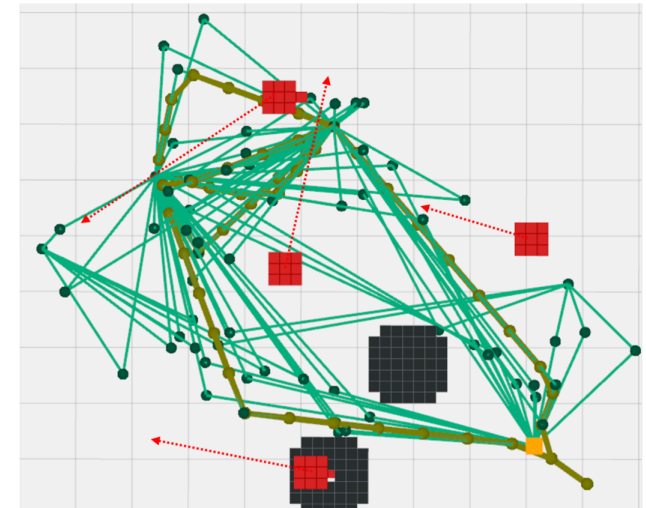
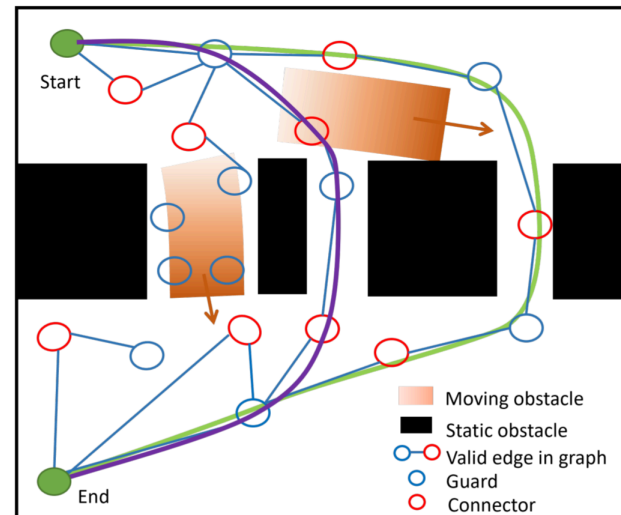
- Start and goal initialized as **Guards**
- Sampled v becomes **Connector** only if visible

Guards g_1, g_2 satisfy $SI(g_1, v) \cap SI(v, g_2) \neq \emptyset$

UTVD-guided insertion

- Same UTVD class + shorter: **replace** existing Connector
- Different UTVD class: **add** as new Connector (new topological branch)
- Both $\rightarrow \pi$ and $\leftarrow \pi$ checked for complete coverage

Result Graph captures distinct **spatiotemporal** topologies, not just geometric paths



Method

B-Spline Optimization: All Four Cost Terms

Control point derivatives (B-spline, knot span t_s)

$$V_i = \frac{Q_{i+1} - Q_i}{t_s}, \quad A_i = \frac{V_{i+1} - V_i}{t_s}, \quad J_i = \frac{A_{i+1} - A_i}{t_s}$$

Full objective $\min_{Q,t} \sum_{d \in \{c, od, ct, f\}} \lambda_d J_d(Q, t)$

Control cost (jerk): $J_c = \sum_i \|J_i\|^2$

Dynamic feasibility

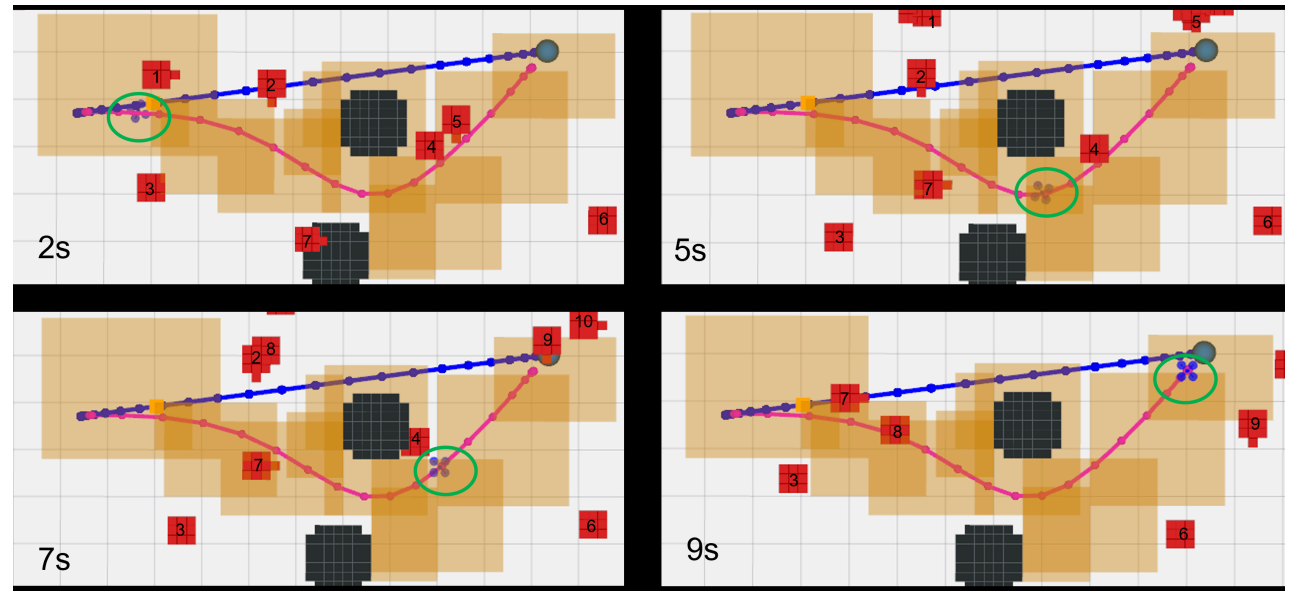
$$J_f = \sum_i \|V_i - v_m\|^2 + \sum_i \|A_i - a_m\|^2$$

Obstacle margin $J_{od} = \sum_i \sum_j J_{ij}$, $d = \|E_j^{-1}(p_i - o_j)\|$

$$J_{ij} = \begin{cases} 0 & \text{if } d > d_{th} \\ (d - d_{th})^2 & \text{if } d \leq d_{th} \end{cases}$$

Corridor — L1 deviation from cuboid

$$J_{ct} = \sum_i \left(\|b_{l,j} - Q_i\|_1 + \|Q_i - b_{u,j}\|_1 \right)$$



| Experiment

Protocol: Three Density Levels, Seven Front-End Planners, Four Metrics

Benchmark setup

- 100 trials per map level
- Map re-generated every 3 trials, random start and goal
- Success = find a collision-free trajectory to goal

Front-end planners compared

- **SIMP** (proposed)
- T-PRM, T-PRM(dyn)
- VIS-PRM, VIS-PRM(dyn)
- SIPP, SIPP-IP

(dyn) variants: add kinodynamic check via trapezoidal velocity profile to the original planner

Map level	Density range	Dynamic obstacles
Sparse	[0, 0.01]	[0, 20]
Moderate	[0.05, 0.1]	[20, 40]
Dense	[0.15, 0.2]	[40, 60]

100

trials per level

3

density levels

4

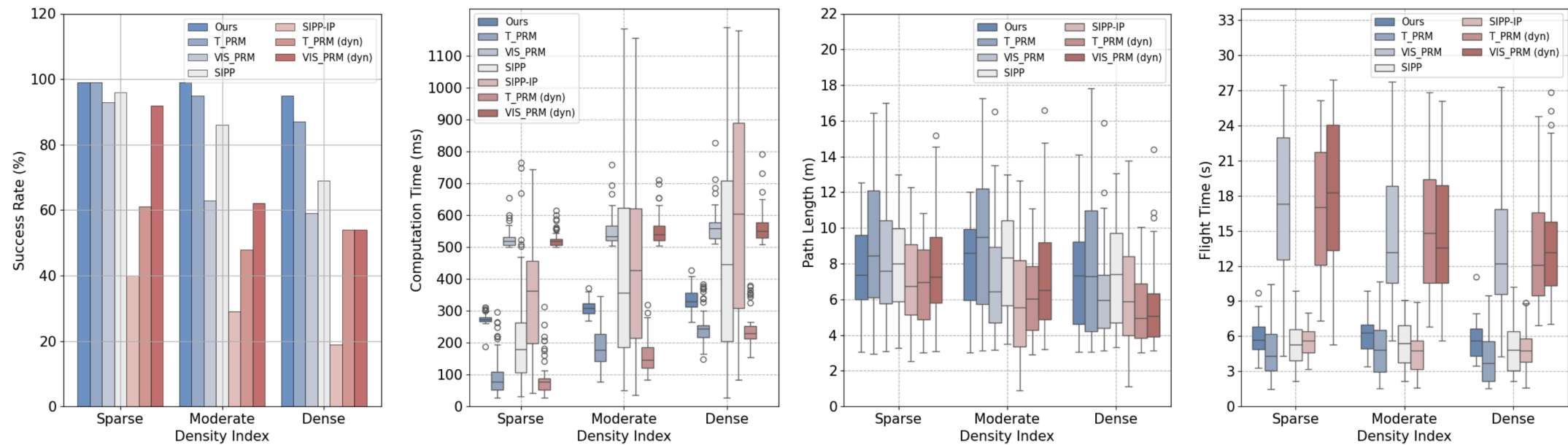
evaluation metrics

3

trials per map

Experiment

Fig. 6 Front End Benchmark Across Three Map Levels



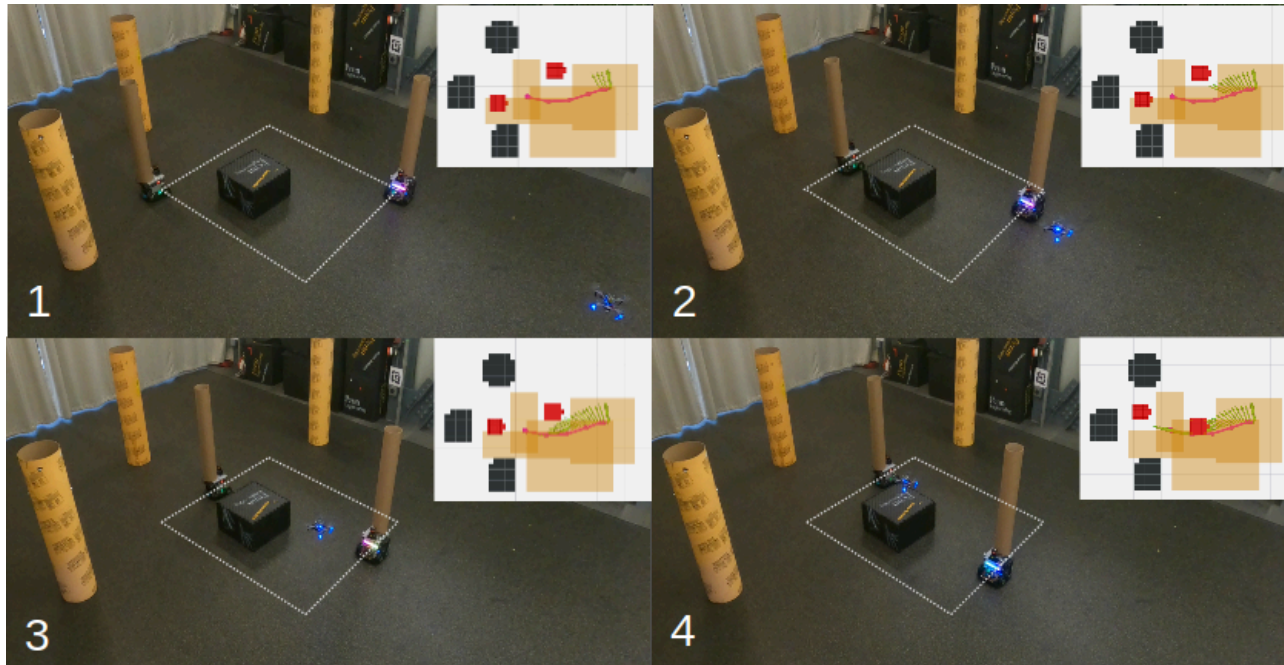
| Experiment

TABLE I
PLANNER COMPARISON

Env.	Methods	Opt. Succ. Rate	Traj. Len. (m)	Flight Time (s)	Ctrl. Cost Avg. (m ² /s ⁵)
Sparse	Ours	100%	7.66	6.04	17.71
	TPRMO	99%	7.89	5.55	72.67
Moderate	Ours	98%	7.93	6.05	26.20
	TPRMO	97%	8.11	5.72	73.27
Dense	Ours	97%	7.44	5.69	30.53
	TPRMO	97%	7.52	5.36	70.31

| Experiment

Hardware Validation: Full Pipeline on a Real Quadrotor



Platform

- **Dragonfly 230**; 2 × **Scarab** as moving obstacles;
- 3 static cylinders
- **Vicon**: shared frame + odometry

Purpose of Vicon

Simulation provides these for free; hardware cannot:

- **Quadrotor pose** — replaces onboard localization
- **Scarab trajectories** — lets planner build safe intervals
- Remove Vicon → no safe intervals → **pipeline breaks**

| Conclusion

Temporal Safety Belongs on Edges, Not Vertices

Conceptual shift

Moving the unit of temporal safety from vertex to edge is the single design choice that makes edge safe intervals, UTVD, and 4D corridors **composable by construction**.

What is actually proven

Representation expressiveness — that capturing spatiotemporal topology this way yields reliable plans.

Trade-off profile

Success advantage largest in **dense scenes**.
SIMP trades time-optimality for structural smoothness.

The real open problem

Vicon **bypasses perception**. When obstacle trajectory predictions are noisy or delayed, how edge safe interval errors propagate to safety is untested.